

1o CHECKPOINT 2o SEMESTRE - PROJETO DIMDIM

Containerizacao de App e Banco em Nuvem no Formato PaaS (ACR & ACI)

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LINKS OFICIAIS DE ENTREGA:

Repositorio GitHub: <https://github.com/Gabriel-Maciel06/devops-cp1-dimdim-azure>

Video de Demonstracao (YouTube): https://youtu.be/sOJWAZk0AmU?is=WEzSpdxw0hyJ5_0e

Swagger UI (Nuvem Azure): <http://app-dimdim-rm562795.chilecentral.azurecontainer.io:8080/swagger-ui.html>

RESUMO DOS RECURSOS AZURE (PaaS):

- Resource Group: rg-dimdim-rm562795 (Regiao: chilecentral)
- Azure Container Registry (ACR): acrdimdim562795.azurecr.io
- Imagem DB: acrdimdim562795.azurecr.io/rm562795-dimdim-db:latest
- Imagem App: acrdimdim562795.azurecr.io/rm562795-dimdim-app:latest (Usuario Non-Root)
- ACI Banco de Dados: rm562795-dimdim-db (PostgreSQL 16 com Azure File Share)
- ACI Aplicacao: rm562795-dimdim-app (Java 21 / Spring Boot 3.3)
- Storage Account: stdimdim562795 (Share: db-dimdim-share montado em /var/lib/postgresql/data)

1. SCRIPT DDL DAS TABELAS (PostgreSQL)

```
CREATE TABLE TB_DIMDIM_TRANSACOES (  
    ID BIGSERIAL PRIMARY KEY,  
    DESCRICAO VARCHAR(150) NOT NULL,  
    VALOR NUMERIC(12, 2) NOT NULL,  
    TIPO VARCHAR(10) NOT NULL,  
    CATEGORIA VARCHAR(50) NOT NULL,  
    DATA_TRANSACAO TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    CONSTRAINT CK_VALOR_POSITIVO CHECK (VALOR > 0),  
    CONSTRAINT CK_TIPO_VALIDO CHECK (TIPO IN ('ENTRADA', 'SAIDA'))  
);
```

2. HOW-TO: COMANDOS AZURE CLI DE PROVISIONAMENTO

1. Criar Grupo de Recursos e Storage Account com File Share

```
az group create --name rg-dimdim-rm562795 --location chilecentral  
az storage account create --name stdimdim562795 --resource-group rg-dimdim-rm562795 --sku Standard_LRS  
az storage share create --name db-dimdim-share --account-name stdimdim562795
```

2. Criar ACR, Realizar Build e Push das Imagens

```
az acr create --resource-group rg-dimdim-rm562795 --name acrdimdim562795 --sku Basic --admin-enabled true  
az acr login --name acrdimdim562795  
docker build -t acrdimdim562795.azurecr.io/rm562795-dimdim-db:latest ./db  
docker push acrdimdim562795.azurecr.io/rm562795-dimdim-db:latest  
docker build -t acrdimdim562795.azurecr.io/rm562795-dimdim-app:latest ./app  
docker push acrdimdim562795.azurecr.io/rm562795-dimdim-app:latest
```

3. Criar ACI do Banco de Dados com Persistencia no Azure Files

```
az container create --resource-group rg-dimdim-rm562795 --name rm562795-dimdim-db \  
--image acrdimdim562795.azurecr.io/rm562795-dimdim-db:latest --ports 5432 --ip-address Public \  
--dns-name-label db-dimdim-rm562795 --azure-file-volume-share-name db-dimdim-share ...
```

4. Criar ACI da Aplicacao DimDim (Non-root user)

```
az container create --resource-group rg-dimdim-rm562795 --name rm562795-dimdim-app \  
--image acrdimdim562795.azurecr.io/rm562795-dimdim-app:latest --ports 8080 --ip-address Public \  
--dns-name-label app-dimdim-rm562795 --environment-variables SPRING_DATASOURCE_URL=... ...
```